

Cloud Computing Course – Assignment 2

Section A – Theory

Q1. Define Cloud Computing in your own words.

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet, allowing users to access resources on demand without owning physical infrastructure.

Q2. Differentiate between IaaS, PaaS and SaaS with one example each.

Service Model	Description	Example
IaaS	Provides virtualized computing resources such as VMs, storage and networking.	Google Compute Engine
PaaS	Provides a platform for developing and deploying applications without managing infrastructure.	Google App Engine
SaaS	Delivers software applications over the internet.	Google Docs

Q3. Explain Region and Zone. Why are multiple zones important?

Region is a geographical area containing multiple data centers. A Zone is an isolated deployment area within a region.

Multiple zones improve availability and fault tolerance. If one zone becomes unavailable, services can continue operating from another zone.

Q4. Explain IAM and the difference between Viewer, Editor and Owner roles.

IAM (Identity and Access Management) controls who can access cloud resources and what actions they can perform.

Viewer: Can view resources only.

Editor: Can view and modify resources.

Owner: Has complete control including IAM permission management.

Section B – Compute Engine & Networking

Q5. Create a Compute Engine VM.

The image shows two screenshots. The top screenshot is from the Google Cloud Platform 'VM instances' page. It displays a table with one instance named 'instance-parth' in the 'us-central1-a' zone. The instance has an internal IP of 10.128.0.4 and an external IP of 34.71.86.112. The bottom screenshot is an SSH terminal window titled 'SSH-in-browser' showing the Debian GNU/Linux 6.1.174-1 prompt. The terminal output includes the Linux kernel version, SMP preemptive dynamic scheduling, and the Debian GNU/Linux version. The prompt is 'parthgulbhile1@instance-parth:~\$'.

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	instance-parth	us-central1-a			10.128.0.4 (nic0)	34.71.86.112 (nic0)	SSH

```
Linux instance-parth 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
parthgulbhile1@instance-parth:~$
```

Q6. Record VM Name, Region, Zone, Machine Type, Internal IP and External IP.

VM Name: instance-parth

Region: us-central1

Zone: a

Machine Type: e2-micro (2 vCPUs, 1 GB memory)

Internal IP: 10.128.0.4

External IP: 34.71.86.112

Explanation:

VM Name: Unique identifier of the virtual machine.

Region: Geographic location where resources are deployed.

Zone: Specific deployment area within a region.

Machine Type: Defines CPU and RAM configuration.

Internal IP: Used for communication within the Google Cloud network.

External IP: Used for internet access.

The screenshot displays the Google Cloud console interface for a VM instance named 'instance-parth'. The left sidebar shows the 'Compute Engine' section with 'VM instances' selected. The main panel shows the 'Details' tab for the instance. The 'Basic information' section includes the following data:

Property	Value
Name	instance-parth
Instance ID	3466311217010812
Description	None
Type	Instance
Status	Running
Creation time	Jun 8, 2026, 11:14:52 pm UTC+05:30

Below the basic information, the 'Network interfaces' section is visible, showing a table with the following data:

NIC type	Primary internal IP address	Alias IP ranges	IP stack type	External IP address	Network tier
	10.128.0.4		IPv4	34.71.86.112 (Ephemeral)	Premium

Q7. Create a Network Tag named mysql-server and firewall rule allowing TCP 3306.

Purpose: This allows MySQL database traffic on port 3306.

The screenshot shows the Google Cloud console interface. The top navigation bar includes the Google Cloud logo, the project name 'My First Project', and a search bar. The left sidebar contains a navigation menu with categories like Compute Engine, Virtual machines, Storage, and Network Security. The main content area is divided into two panels. The top panel, titled 'Edit my-vm instance', shows settings for a VM instance named 'my-vm', including network tags ('mysql-server'), storage (boot disk), and firewall policies. The bottom panel, titled 'Firewall policies', displays a table of existing firewall rules.

Firewall policies table:

Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network	Logs
allow-mysql	Ingress	mysql-server	IP ranges:	tcp:3306	Allow	1000	default	Off
https-allowed	Ingress	Apply to all	IP ranges:	tcp:443	Allow	1000	default	Off
default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	65534	default	Off
default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	65534	default	Off

Q8. Create a Network Tag named postgres-server and firewall rule allowing TCP 5432.

Purpose: This allows PostgreSQL database traffic on port 5432.

The screenshot displays the Google Cloud console interface. The top navigation bar shows the 'Compute Engine' section. The left sidebar lists various services, including 'Virtual machines', 'Storage', 'Network Security', and 'Cloud Armor'. The main content area is divided into two panels. The top panel, titled 'Edit my-vm instance', shows configuration details for a VM instance named 'my-vm'. It includes sections for 'Network tags' (listing 'mysql-server' and 'allow-postgres'), 'Storage' (showing a boot disk named 'my-vm' with a size of 10 GB), and 'Deletion rule' (set to 'Keep disk'). The bottom panel, titled 'Firewall policies', shows a list of firewall rules. A warning message at the top indicates that the user does not have the required permissions to view the inherited firewall policies. Below this, a table lists the firewall rules:

Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network	Logs
allow-mysql	Ingress	mysql-server	IP ranges:	tcp:3306	Allow	1000	default	Off
allow-postgres	Ingress	postgres-server	IP ranges:	tcp:5432	Allow	1000	default	Off
https-allowed	Ingress	Apply to all	IP ranges:	tcp:443	Allow	1000	default	Off

Q9. Explain the purpose of firewall rules and network tags in GCP.

Firewall Rules

Firewall rules control incoming and outgoing network traffic to GCP resources, helping secure VMs by allowing or denying specific ports and protocols.

Network Tags

Network tags are labels assigned to VM instances that allow firewall rules and other network policies to be applied only to selected VMs.

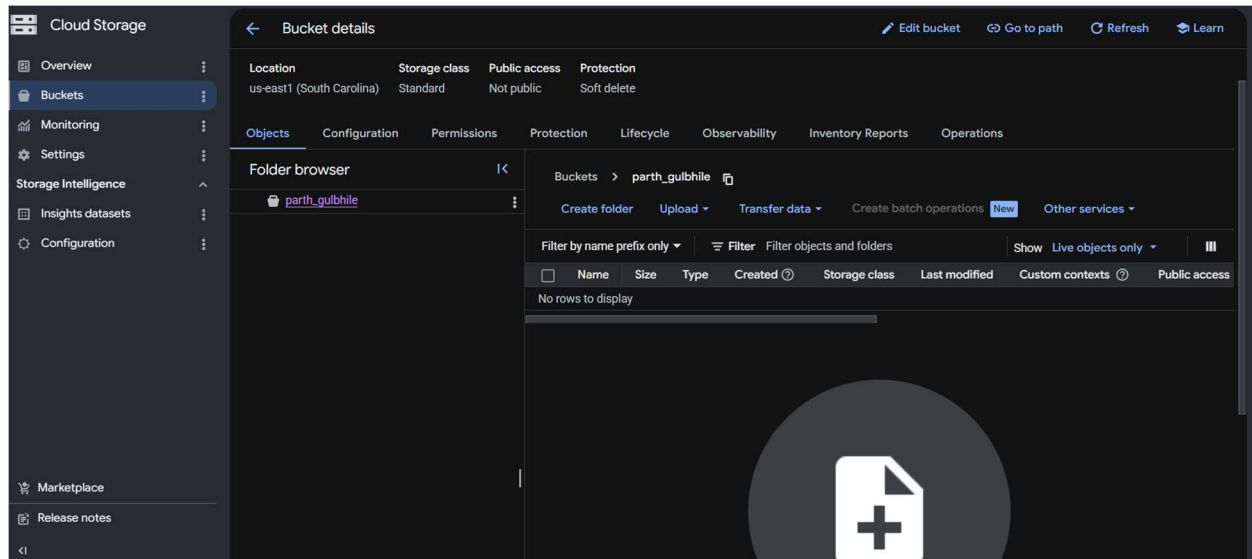
Example

- VM with tag mysql-server receives traffic on port **3306**.
- VM with tag postgres-server receives traffic on port **5432**.
- Other VMs remain unaffected.

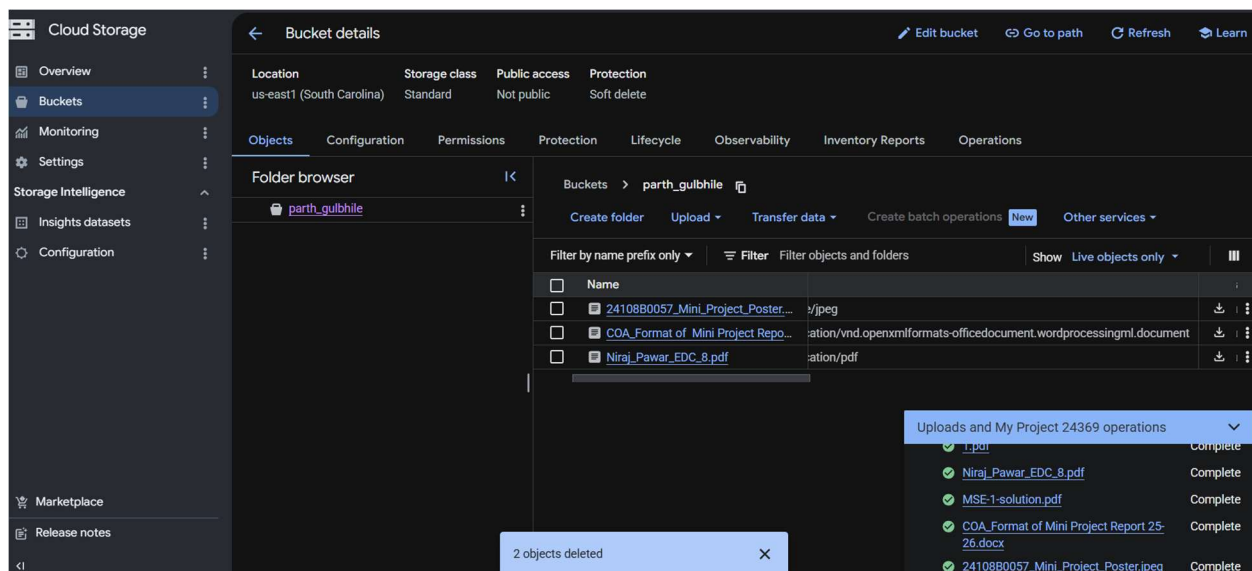
Section C – Cloud Storage (10 Marks)

Q10. Create a Cloud Storage Bucket.

Insert:



Q11. Upload at least three files of different types.



Q12. Explain the difference between Object Storage and Traditional File Systems.

Object Storage

Stores data as objects along with metadata and a unique identifier.

Traditional File System

Stores data as files organized in folders and directories.

Object Storage

Designed for massive scalability in cloud environments.

Accessed using APIs, URLs, or cloud services.

Example: Google Cloud Storage.

Traditional File System

Designed mainly for local disks and operating systems.

Accessed using file paths and operating system commands.

Example: NTFS (Windows), ext4 (Linux).

Q13. Record Bucket Name, Region and Storage Class.

Bucket Name: parth_gulbhile

Region: us-east1 (South Carolina)

Storage Class: Standard

Explanation:

Bucket Name: Unique identifier for the storage bucket.

Region: Location where bucket data is stored.

Storage Class: Defines cost and availability characteristics.

Section D – Pub/Sub (5 Marks)

Q14. Create a Topic and Subscription.

The screenshot displays the Google Cloud Pub/Sub console interface. The left sidebar shows the navigation menu with 'Topics' and 'Subscriptions' highlighted. The main content area is split into two sections: 'Topics' and 'Subscriptions'.

Topics Section:

- Header: 'Topics' with buttons for 'Create topic', 'Delete', and 'Tags'.
- Card: 'Pub/Sub offers SMTs to simplify data transformations for streaming pipelines'. It describes performing single message transforms like casting, redaction, masking, complex filtering, to message attributes and/or the data directly within Pub/Sub using new JavaScript User-Defined Functions (UDFs). A 'Learn more' link is present.
- Table: A table with columns 'Topic ID', 'Encryption', and 'Ingestion source'. One row is visible with 'my-topic' as the Topic ID and 'Google-managed' as the Encryption. A dropdown menu is open showing 'projects/project-8da5664d4462-40e9-8c0/topics/my-topic'.
- Right Panel: 'Select a topic' with tabs for 'Permissions', 'Labels', and 'Storage policy'. A message says 'Please select at least one resource.'

Subscriptions Section:

- Header: 'Subscriptions' with buttons for 'Create subscription', 'Delete', and 'Tags'.
- Card: 'Pub/Sub offers SMTs to simplify data transformations for streaming pipelines'. It describes performing single message transforms such as casting, redaction, masking, complex filtering, to message attributes and/or the data directly within Pub/Sub using new JavaScript User-Defined Functions (UDFs). A 'Learn more' link is present.
- Table: A table with columns 'State', 'Subscription ID', 'Delivery type', 'Topic name', and 'Ack deadline'. One row is visible with 'my-subscription' as the Subscription ID, 'Pull' as the Delivery type, and '10 seconds' as the Ack deadline.
- Right Panel: 'Select a subscription' with tabs for 'Permissions' and 'Labels'. A message says 'Please select at least one resource.'

Q15. Publish a message and retrieve it using Subscription.

The screenshot shows the Google Cloud Pub/Sub 'Publish' page for a topic named 'my-topic'. The left sidebar contains navigation links: Pub/Sub, Overview, Topics (selected), Subscriptions, Snapshots, Schemas, Pub/Sub Lite, Lite Reservations, Lite Topics, and Lite Subscriptions. The main content area is divided into two panels. The left panel, titled 'my-topic', shows an 'Export options have moved' message and a 'Publish message' button. The right panel, titled 'Publish count', contains fields for 'Number of messages' (set to 1), 'Message interval (seconds)' (set to 1), and a 'Message body' text area containing 'Hello Pub/Sub'. Below the message body is a 'Message attributes' section with a warning about attribute limits. At the bottom of the right panel are 'Publish' and 'Cancel' buttons.

Google Cloud My First Project

Pub/Sub / Topics / Topic: my-topic

Pub/Sub Overview Topics Subscriptions Snapshots Schemas Pub/Sub Lite Lite Reservations Lite Topics Lite Subscriptions

my-topic Edit

Export options have moved Got it

Topic name Tags

Subscriptions Snapshots

Test if your subscription can receive a message

Step 1

Manually publish a message containing data and attributes with metadata to this topic.

Publish message Manually publish a message

Step 2

Select a Cloud Pub/Sub subscription to pull messages from

Select a Cloud Pub/Sub Subscription

Publish count

You can publish the given message once or multiple times in an interval. This can be useful for getting messages in new subscriptions and testing. For a more robust way to publish messages multiple times, consider using Cloud Scheduler.

Number of messages * 1

Enter an amount between 1-100.

Message interval (seconds) 1

How long to wait before publishing the next message

Message body

The message you want to publish to this topic. Either message or attribute will be required to publish.

Message Hello Pub/Sub

Message size should not exceed 10MB.

Message attributes

At most 100 attributes per message, attribute key should not start with 'goog' and should not exceed 256 bytes, attribute value should not exceed 1024 bytes.

Publish Cancel

on receive a manually published message

the containing data and attributes with metadata to this topic

Manually publish a message containing data and attributes with metadata to this topic.

Subscription to pull messages from

Subscription to pull messages from

Message published.



Section E – BigQuery (10 Marks)

Q16. Create a Dataset in BigQuery.

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains a navigation menu with options like Queries, Notebooks, Data canvases, Data preparations, Pipelines, Connections, and my_dataset. The main area is titled 'Create new' and features buttons for SQL query, Notebook, Notebook with Spark, ML model, Graph, Agent, Conversation, and Pipeline. Below this, there are buttons for Data preparation, Data canvas, and Table. A 'Recent' section displays a table with the following data:

Display name	Type	Last modified time	Project
my_dataset	Dataset	Jun 8, 2026, 10:25:14 PM	project-8da5664d-f462-40e9-8c0
Hevo_temp_bi_usa_daily_trends	Table	Jun 8, 2026, 5:41:50 PM	bigquery-public-data
dataflix_covid	Dataset	Jun 8, 2026, 5:41:47 PM	bigquery-public-data
google_trends	Dataset	Jun 8, 2026, 5:41:40 PM	bigquery-public-data

At the bottom, there is a 'Try with templates' section with three options: 'Try the Google Trends Demo Query', 'Try the Colab Demo Notebook', and 'Try the Spark Demo Notebook'.

Q17. Upload a CSV file into a table.

The screenshot shows the Google Cloud BigQuery Studio interface. On the left is a navigation sidebar with options like Overview, Studio, Agents, Pipelines & Integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, and Administration. The main panel displays the 'my_table' schema for the dataset 'my_dataset' in project 'project-8da5664d-f462-40e9-8c0'. A message at the top suggests generating table and column descriptions with Gemini. Below this, a table lists the schema fields:

Field name	Type	Mode	Description	Key	Collation	Default Value	Policy Tags
id	INTEGER	NULLABLE					
name	STRING	NULLABLE					
city	STRING	NULLABLE					

Buttons for 'Edit schema' and 'View row access policies' are visible at the bottom of the schema view.

Q18. Run the following queries:

The screenshot shows the Google Cloud BigQuery Studio interface with a query being executed. The query is:

```
1 SELECT * FROM `project-8da5664d-f462-40e9-8c0.my_dataset.my_table`  
2 LIMIT 5;
```

The query results are displayed in a table with 5 rows:

Row	id	name	city
1	1	Sujal	Mumbai
2	2	Rahul	Pune
3	3	Priya	Nashik
4	4	Amit	Nagpur
5	5	Neha	Delhi

The interface also shows a status message: "This query will process 110 B when run. Using on-demand processing quota."

The screenshot shows the Google Cloud BigQuery Studio interface with a second query being executed. The query is:

```
1 SELECT COUNT(*) AS total_rows  
2 FROM `project-8da5664d-f462-40e9-8c0.my_dataset.my_table`;
```

The query results are displayed in a table with 1 row:

Row	total_rows
1	5

The interface also shows a status message: "Query completed".

Q19. Explain the difference between a Dataset and a Table in BigQuery.

Dataset

A dataset is a container used to organize and manage tables in BigQuery.

A dataset can contain multiple tables.

Used for logical grouping of data.

Example: my_dataset

Table

A table is a collection of rows and columns that stores actual data.

A table belongs to a single dataset.

Used for storing and querying data records.

Example: my_table

Bonus Challenge (10 Marks)

Cloud Architecture for an Online Food Delivery Application

Flow:

Customer App → Compute Engine → Pub/Sub → Restaurant/Delivery Services

Customer App → Compute Engine → Cloud Storage

Compute Engine → BigQuery → Analytics Dashboard

Complete Flow: Customer Order to Analytics Dashboard

1. Customer Places Order

The customer uses a mobile app or website to browse restaurants and place a food order.

2. Compute Engine

The food delivery application runs on **Compute Engine**. It handles:

- User login
- Restaurant menu display
- Order processing
- Payment handling
- Order status updates

3. Cloud Storage

Cloud Storage stores:

- Food images

- Restaurant menu images
- Invoices and receipts
- Application documents

4. Pub/Sub

When an order is placed:

- The application publishes an order event to Pub/Sub.
- Pub/Sub distributes messages to different services.
- Notifications can be sent to restaurants, delivery partners, and customers.

5. BigQuery

Order data from Pub/Sub is stored in BigQuery for analysis:

- Total orders
- Revenue
- Popular food items
- Customer trends
- Delivery performance

6. Analytics Dashboard

Business users view dashboards created from BigQuery data to monitor:

- Daily sales
- Monthly revenue
- Top restaurants
- Most ordered dishes
- Customer growth

Role of Each Service

Service	Role
Compute Engine	Hosts the food delivery web/mobile backend application.

Service	Role
Cloud Storage	Stores food images, menus, invoices, and other files.
Pub/Sub	Provides real-time messaging and event distribution for orders and notifications.
BigQuery	Stores and analyzes order data for reporting and business intelligence.
Customer App	Interface used by customers to place orders.
Analytics Dashboard	Displays reports and insights generated from BigQuery data.

Conclusion

The architecture uses **Compute Engine** for application hosting, **Cloud Storage** for file storage, **Pub/Sub** for real-time event processing, and **BigQuery** for analytics. This enables a scalable and efficient online food delivery platform with real-time order processing and business insights.